PRIMEX FCoEE

RoboCupJunior Rescue Maze 2026

INCHEON RoboCup 2026

AFIF (LEADER) – HARDWARE AND MOVEMENT LEAD

HAZIEQ – GREEK LETTER DETECTION

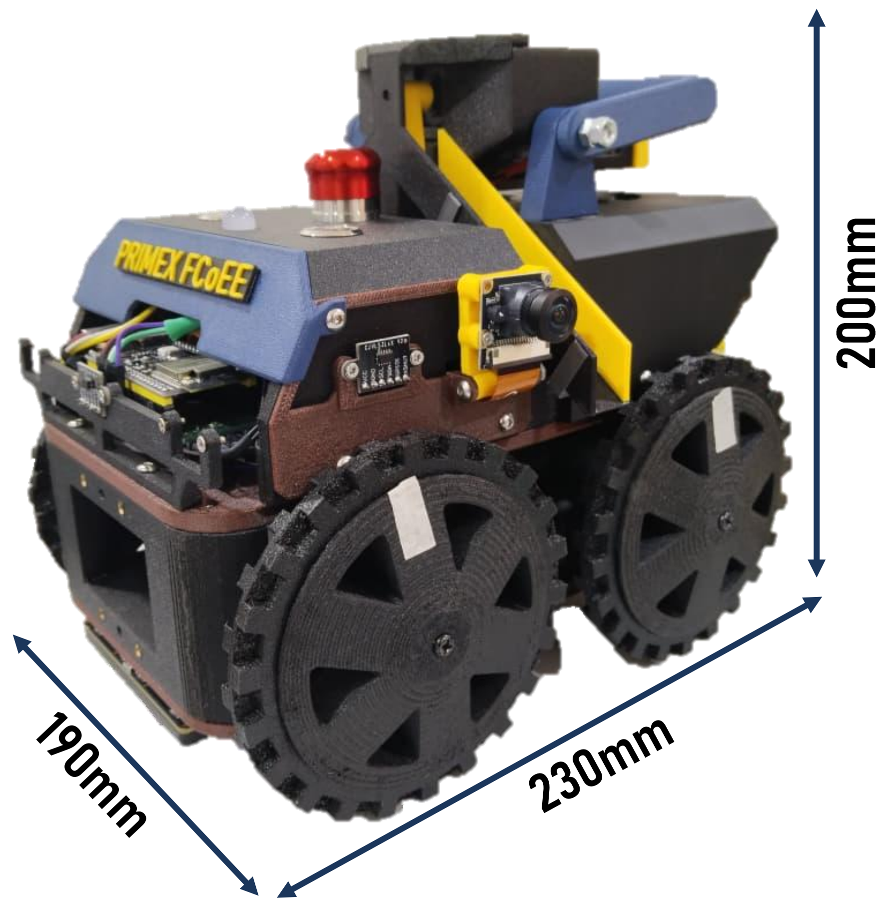
MUQRI (MENTOR 1) – CONSULT–DESIGN & HARDWARE

HAFIY(CO-LEADER) – MAPPING AND NAVIGATION LEAD

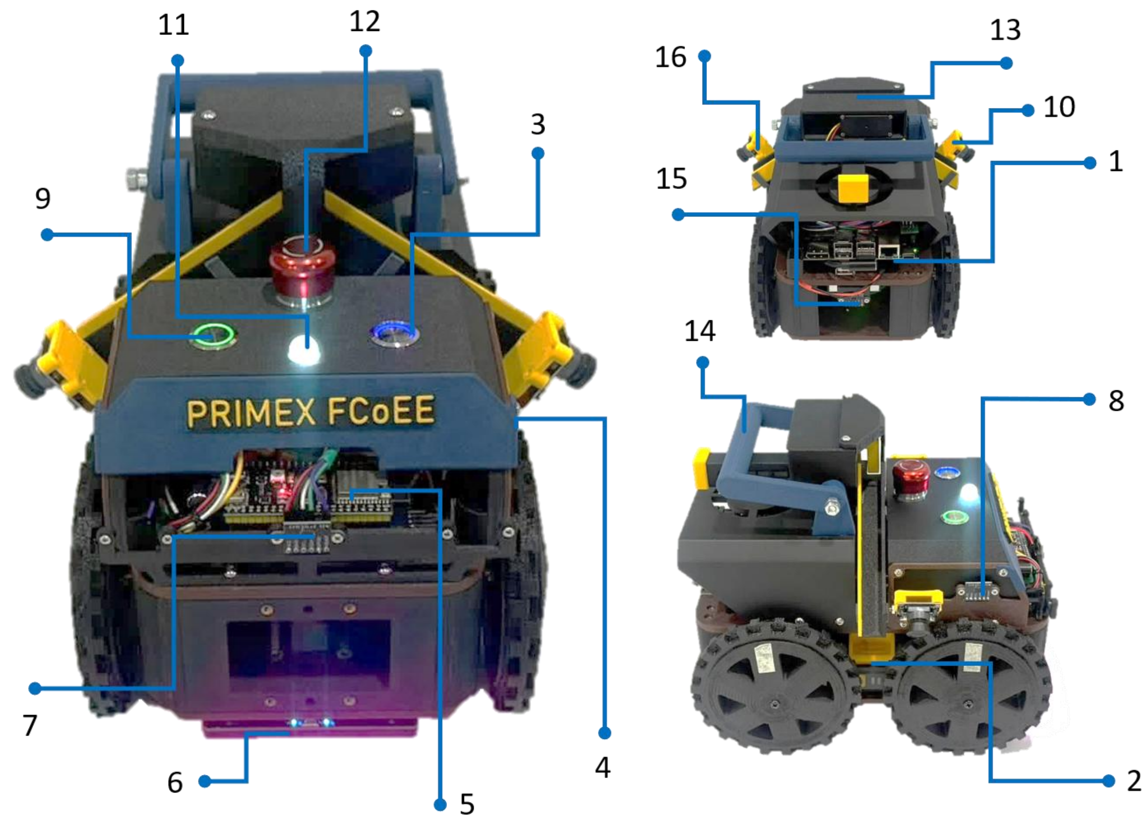
ADAM – COGNITIVE TARGET DETECTION

AMEERUL (MENTOR 2) – CONSULT–ALGORITHM & STRATEGY

1 HARDWARE PART



COMPONENT CONFIGURATION



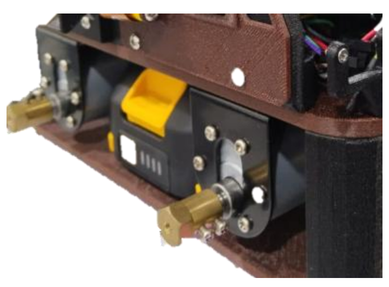
No	Component
1	Jetson Orin Nano
2	20V 1.5Ah Battery
3	Power Button
4	SL Distance Sensor (VL53L0X)
5	ESP32 S3 Devkit Board
6	mBlock Colour Sensor
7	FC Distance Sensor (VL53L0X)
8	SR Distance Sensor (VL53L0X)
9	Start Button
10	Right Side IMX219 Camera
11	RGB LED Indicator
12	Stop Button
13	MG996R Servo + Drop Kit Mechanism
14	Main Handle
15	RC Distance Sensor (VL53L0X)
16	Left Side IMX219 Camera

HARDWARE INNOVATION

CF-REINFORCE MATERIAL FOR CHASSIS

ABS-CF for body panels (high stiffness, heat resistance), PETG-CF for chassis frame (toughness under drive loads), PLA for accessories.

CF materials reduce flex under field impact.



HORIZONTAL SPRING-LOADED RESCUE KIT DEPLOYMENT

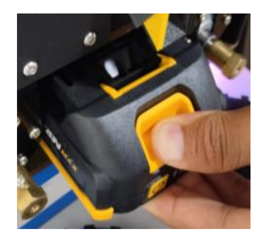
Spring-loaded magazine with dual slide ramps (left and right). MG996R servo actuates the ejector arm to deploy kit within 15cm of the victim.

DUAL CAMERA ADJUSTABLE ANGLE

Two IMX219 160° wide-angle cameras mounted left and right at 100–110mm height with adjustable 15–20° tilt. Enables continuous bilateral victim wall scanning while moving — no stop-and-rotate needed. Tilt mechanism allows field-day angle tuning.

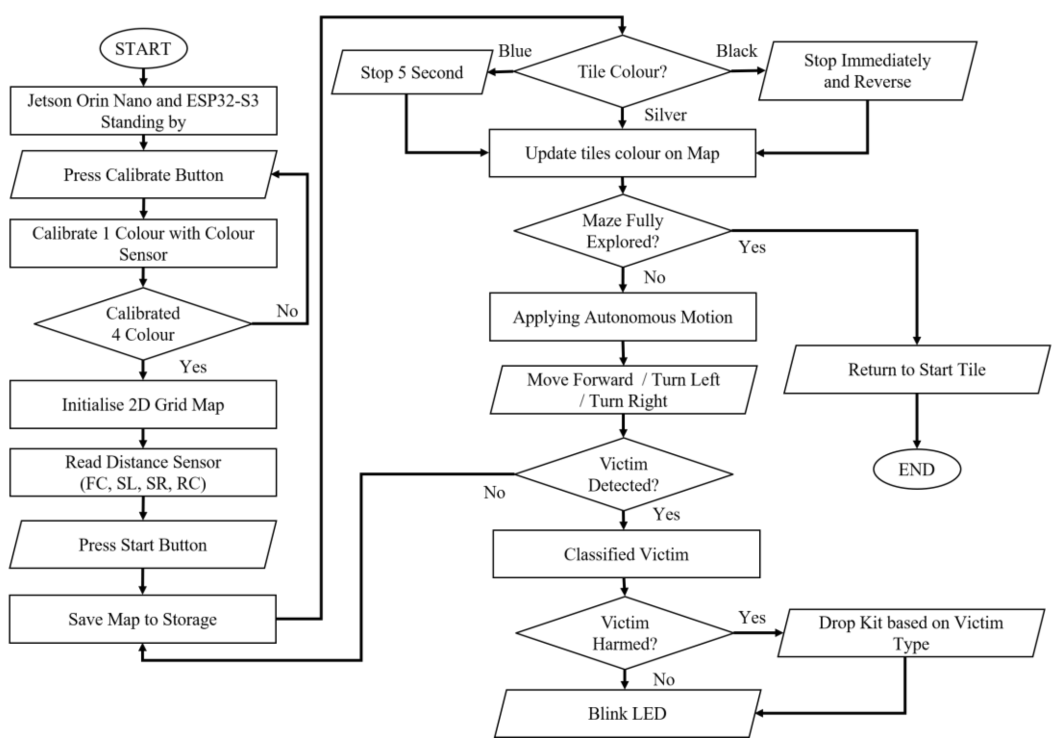
SWAPPABLE POWER TOOL BATTERY DOCK

20V 1.5Ah power tool battery on a quick-release dock. Swappable in seconds between rounds without tools ensuring full power capacity for every scoring run.



2 SOFTWARE PART

SYSTEM ARCHITECTURE



The robot calibrates floor colors, initializes the 2D grid map, then navigates autonomously using the Right-Hand Rule. It responds to floor tiles in real time, stops on blue, reverses on black, and saves checkpoints on silver. Both cameras scan for victims simultaneously, deploying rescue kits and blinking the RGB LED on detection.

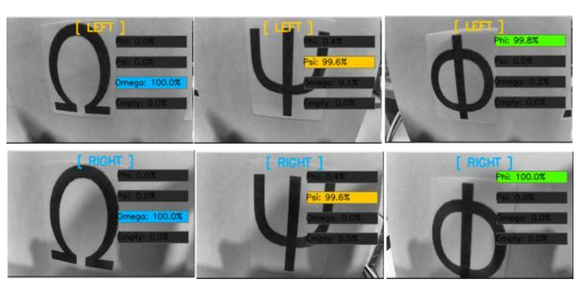
ALGORITHMS & METHODS

GREEK LETTER DETECTION

S1 – Capture Dataset for 10,500 images, 3500 for each letter class.

S2 – Train TensorFlow CNN to generate model.

S3 – Detection based on trained model



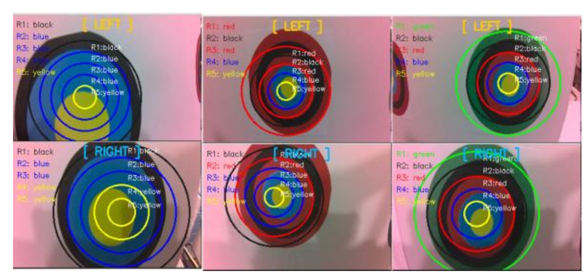
COGNITIVE TARGET DETECTION

S1 – Detect outermost circle

S2 – Template matching into 1:2:3:4:5 diameter ratio

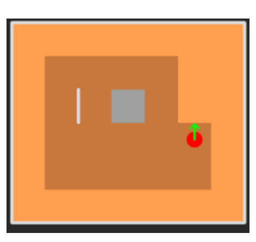
S3 – NumPy masking per ring with HSV dominant color.

S4 – Sum weightage for all 5 rings



AUTONOMOUS MAZE NAVIGATION

RIGHT HAND RULE – Follow wall for exploring and mapping tile.



EMPTY TILE – Recover based on non-discovered tile from mapping